Case Study – 2 (50 Marks)
Consider a two-wheeler manufacturing company that manufactures:

Motorcycles
Scooters
Electric scooters
Electric motorcycles

The company operates multiple manufacturing plants and has a network of suppliers and authorized dealers.

The company currently maintains separate systems for:

- Vehicle models
  - Example :
    - Model Name : X-Sport 160
    - Model Code : XSP160
    - Vehicle Type : Motorcycle
    - Engine Capacity : 160 cc
    - Fuel Type : Petrol
    - Transmission : 5-Speed
    - Seating Capacity : 2
    - Ex-Showroom Price: ₹1,25,000
- Components
  - Example:
    - Engine Battery Brake assembly Tyre Wheel Headlamp ECU
- Suppliers
  - Example:
    - Name, Address, Type, Components { Ex. Brake Assembly, Clutch Assembly, Suspension Components }
- Manufacturing
- Inventory
- Employees
- Dealers
- Customers
- Sales
- Service
- Warranty

Because these systems are not properly integrated, management faces problems such as:

- Difficulty tracking vehicles from manufacturing to customer delivery
- Duplicate customer and vehicle records
- Difficulty tracking components used in a vehicle
- Supplier-performance issues
- Incorrect inventory information
- Difficulty tracking warranty claims
- Difficulty identifying vehicles affected by defective components
- Difficulty tracking service history
- Difficulty calculating production costs

The company therefore wants to develop an integrated Two-Wheeler Manufacturing Database Management System.

1. Develop a E-R Diagram for the aforesaid scenario with all information. **(10 Marks)**
2. Convert the E-R diagram into Relational Schema **(10 Marks)**
3. Implement the Relational schema in the MySQL / PostgreSQL server and insert records in all tables. Note: Additional records may required to answer question no 4. **(10 Marks)**
4. Consider if the Database system was developed using NoSQL, what are the advantages and/or disadvantages exist in comparision with RDBMS. List 10 points **(10 Marks)**
5. What are the references (textbook / notes/ internet sources / LLMs) you used to solve the problem including any LLM such as ChatGPT etc.? What is your satisfactory level of the reference you used to solve the problem? Give a detailed report on a) the prompt you provided or the page you referred, b) the answer you obtained, c) satisfactory level of its answer in 1 – 5 point scale. List atleast 10 references **(10 Marks)**